Procedure of ELISA

2023

A. Antigen Coating

1. Prepare human IgG solution at 0.025 mg/mL in carbonate-bicarbonate buffer (coating buffer).
2. Pipette 0.2 mL of the above solution to each well of the 96-well microtiter plate from 2B-D to 10 B-D. Incubate at 37 ℃ for 30-45 min.
3. Remove the coating solution. Wash three times with PBS-T (0.2 mL/well).
4. Blocking step: pipette 0.02% (*w/v*) non-fat dry milk in-PBST (0.2 mL/well) to each well of the 96-well microtiter plate from 2B-D to 11 B-D except 10 B-D. Keep the plate at Room Temperature for 5-10 min. Wash as mentioned above.

B. Primary Antibody Reaction

5. Dilute the primary antibodies, Rabbit-anti-human IgG, in PBS-T. (Different dilutions of 1°Ab. , 1:400~1:12800).
6. Add 0.2 mL of the diluted first antibody to each well according to Figure 1.
   Note: Add sample A (1st Ab. of unknown concentration) to well 8B-D. Negative control should be included (9B-D, No first Ab.).
7. Incubate at 37 ℃ for 1 h.
8. Wash three times with PBS-T (0.2 mL/well).

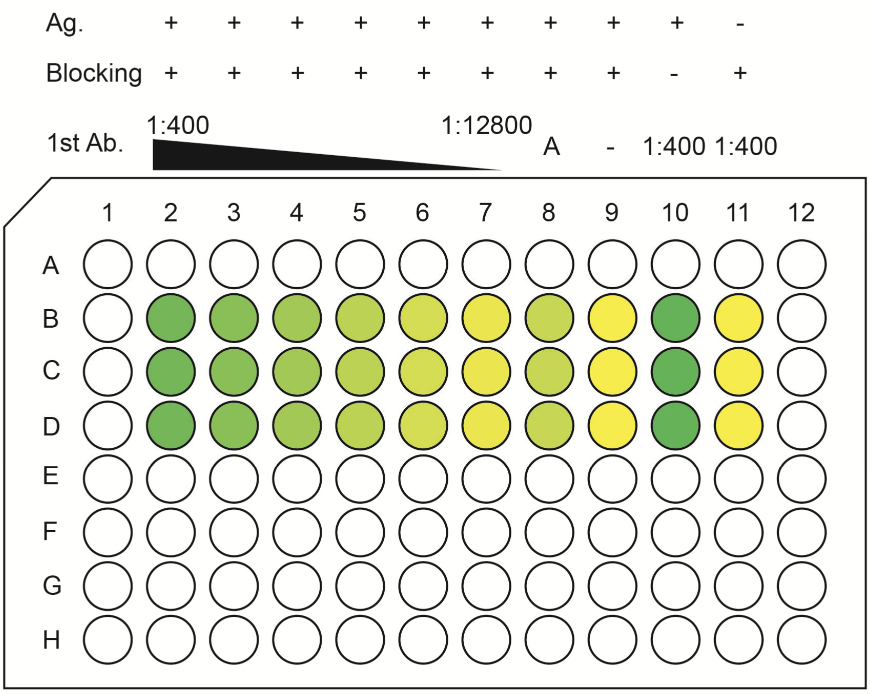

Fig1 ELISA in 96-well microtiter plate

C. Application of Secondary Antibody

9. Dilute (1:40,000-1:20,000) the peroxidase conjugated Goat-anti-rabbit IgG-HRP (secondary antibody, 2°Ab) in PBS-T. Add 0.2 mL: of this solution to each well.
10. Incubate at 37 ℃ for 30 min.
11. Wash three times with PBS-T (0.2 ml/well).

D. Substrate Preparation

12. During the last incubation and immediately before use, dissolve o-Phenylenediamine in Citric acid-sodium hydrogen phosphate buffer; add 30% H_2O_2 before use.

E. Development

13. Add 0.2 mL of the freshly prepared substrate to each well.
14. Orange-yellow color should develop in positive wells after 3-5 min.
15. Stop reaction with 50 µL per well of preferred stopping reagent and read at 490 nm in a microplate reader.

REAGENTS

1. Coating buffer (Carbonate-Bicarbonate buffer, pH 9.6) :
   Na_2CO_3 0.15 g,
   $NaHCO_3$ 0.293 g, add H_2O to 100 mL (pH 9.6).
2. Phosphate buffered saline (PBS):
   NaCl 8.0 g,
   KCl 0.20 g,
   KH_2PO_4 0.24 g,
   $Na_2HPO_4 \cdot 12\ H_2O$ 2.9 g, add H_2O to 1000 mL (pH 7.4).
3. Washing buffer (PBS-T): PBS containing 0.05% Tween 20.
4. Primary antibody: Rabbit-anti-human IgG solution (1 mg/mL; 1:400 diluted 1°Ab. will be distributed to each group).
5. Peroxidase conjugated secondary antibody: Goat-anti-rabbit IgG-HRP (1:40,000-1:20,000).
6. Substrate: substrate buffer (fresh):
   0.1 M Citric acid (2.10 g/100 mL), 6.1mL;
   0.2 M $Na_2HPO_4 \cdot 12\ H_2O$ (7.163 g/100mL), 6.4mL;
   add H_2O 12.5 mL,
   dissolve 8 mg o-Phenylenediamine,
   and add 40 µL of 30% H_2O_2 before use.
7. Stopping reagent: 2 M H_2SO_4.